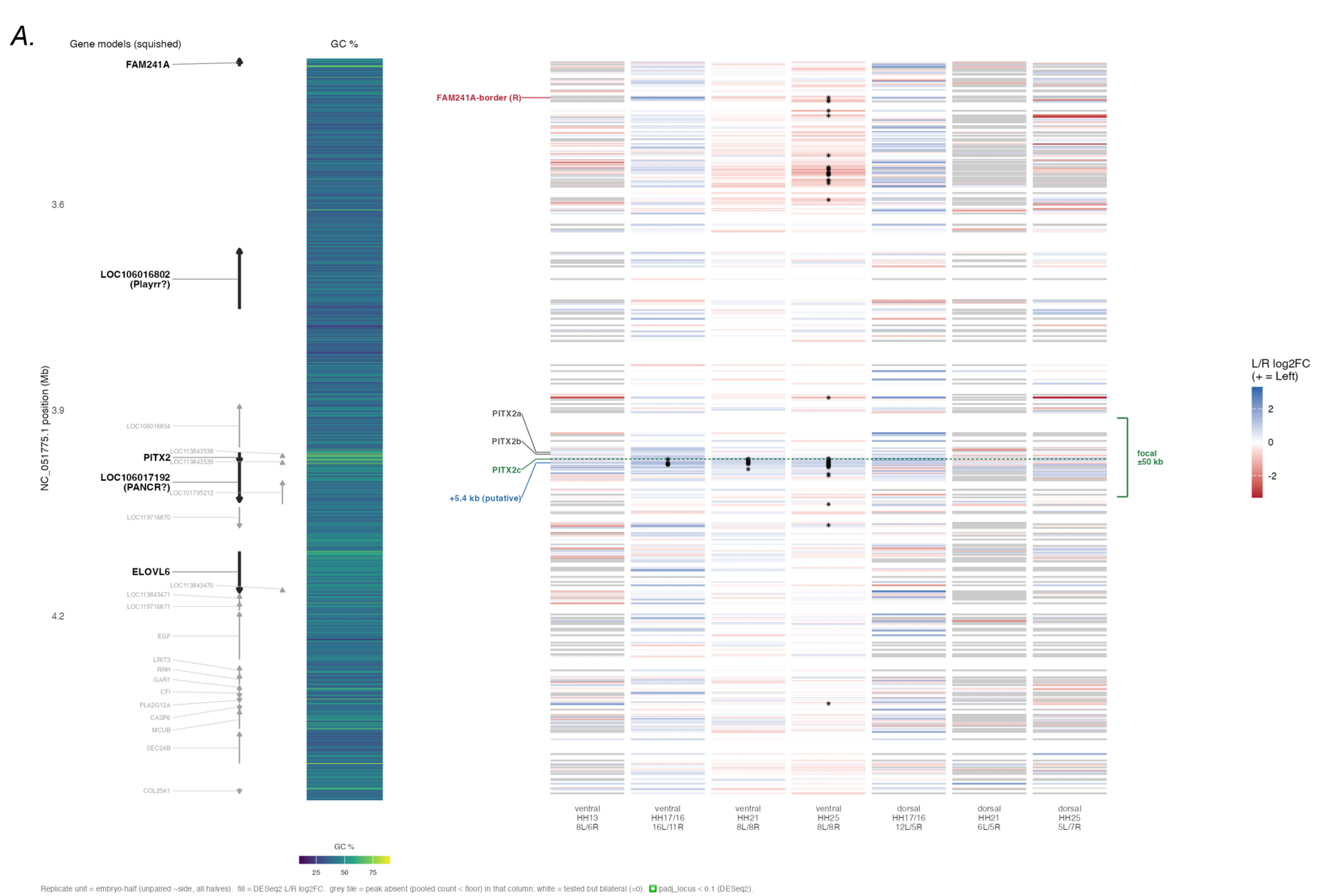
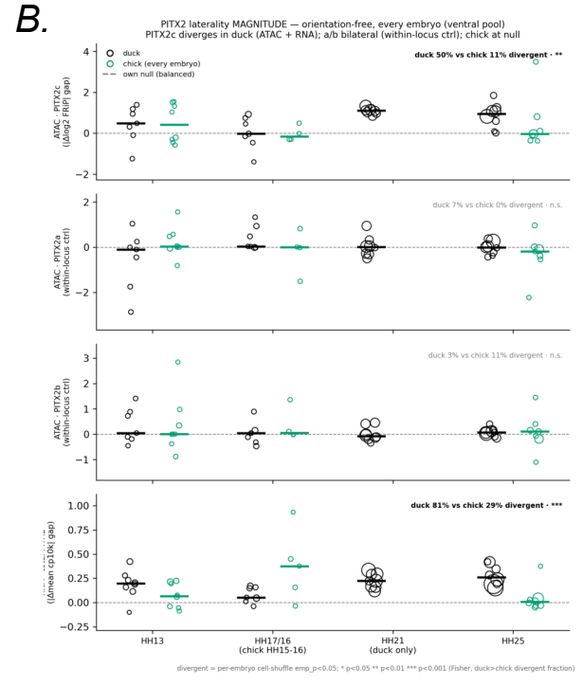
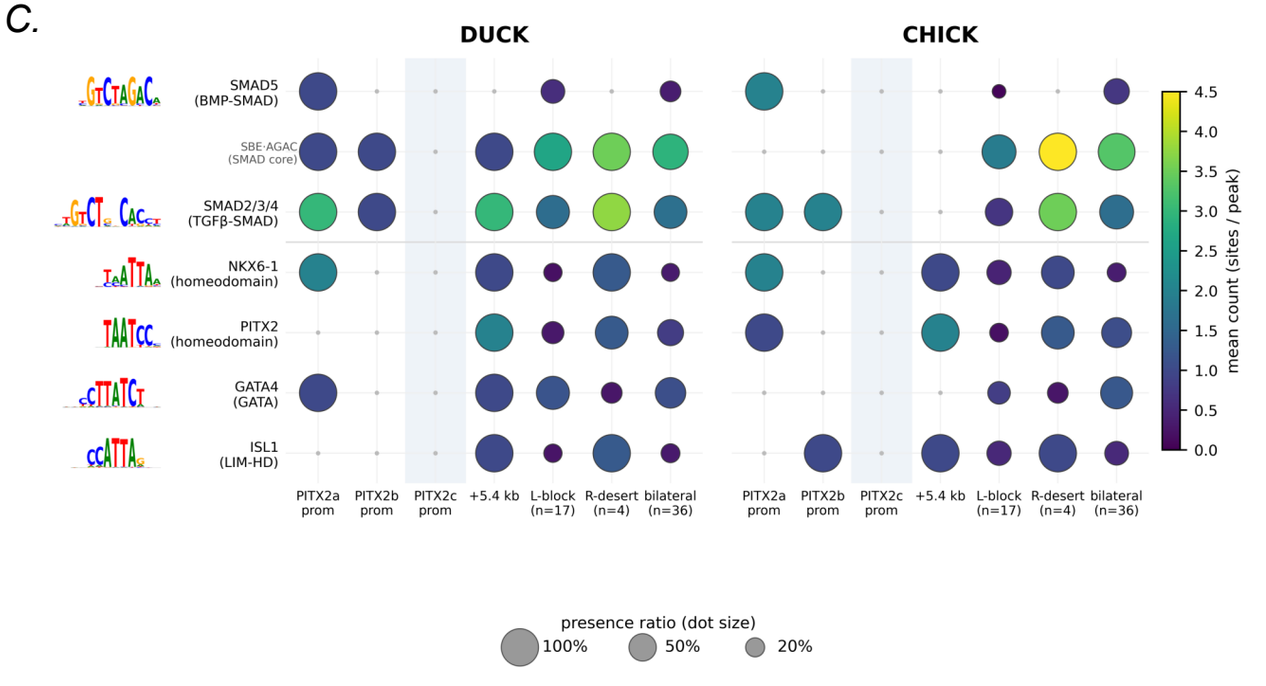

A.
Gene models (squished)
GC %
FAM241A
LOC106016802
(Playrr?)
PITX2
LOC106017192
(PANCR?)
ELOVL6
EGF
LRIT3
RRH
GAR1
CFI
PLA2G12A
CASP6
MCUB
SEC24B
COL25A1
NC_051775.1 position (Mb)
3.6
3.9
4.2
FAM241A-border (R)
PITX2a
PITX2b
PITX2c
+5.4 kb (putative)
focal
±50 kb
L/R log2FC
(+ = Left)
2
0
-2
ventral
HH13
8L/6R
ventral
HH17/16
16L/11R
ventral
HH21
8L/8R
ventral
HH25
8L/8R
dorsal
HH17/16
12L/5R
dorsal
HH21
6L/5R
dorsal
HH25
5L/7R
Replicate unit = embryo-half (unpaired ~side, all halves). fill = DESeq2 L/R log2FC. grey tile = peak absent (pooled count < floor) in that column; white = tested but bilateral (=0). padj_locus < 0.1 (DESeq2).


B.
PITX2 laterality MAGNITUDE — orientation-free, every embryo (ventral pool)
PITX2c diverges in duck (ATAC + RNA); a/b bilateral (within-locus ctrl); chick at null
duck
chick (every embryo)
own null (balanced)
duck 50% vs chick 11% divergent · **
duck 7% vs chick 0% divergent · n.s.
duck 3% vs chick 11% divergent · n.s.
duck 81% vs chick 29% divergent · ***
HH13
HH17/16
(chick HH15-16)
HH21
(duck only)
HH25
divergent = per-embryo cell-shuffle emp_p<0.05; * p<0.05 ** p<0.01 *** p<0.001 (Fisher, duck>chick divergent fraction)


C.
DUCK
CHICK
SMAD5
(BMP-SMAD)
SBE·AGAC
(SMAD core)
SMAD2/3/4
(TGFβ-SMAD)
NKX6-1
(homeodomain)
PITX2
(homeodomain)
GATA4
(GATA)
ISL1
(LIM-HD)
PITX2a prom
PITX2b prom
PITX2c prom
+5.4 kb
L-block (n=17)
R-desert (n=4)
bilateral (n=36)
mean count (sites / peak)
presence ratio (dot size)
100%
50%
20%